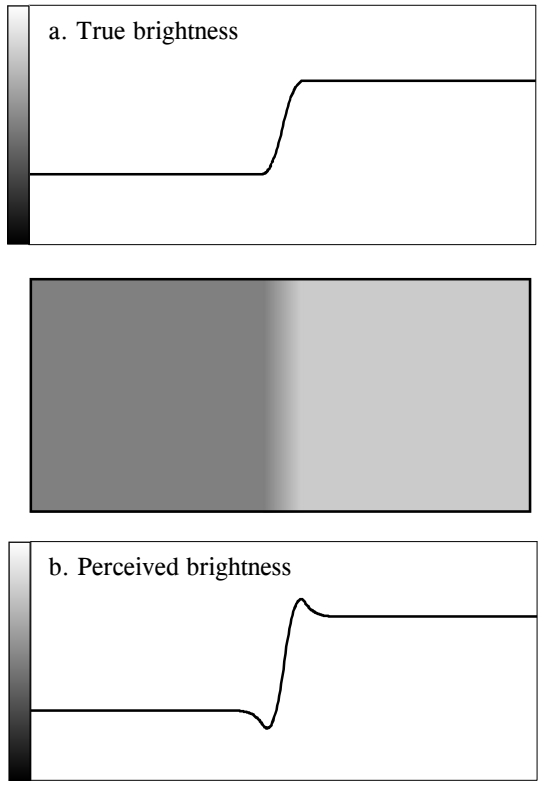


FIGURE 24-2
Mach bands. Image processing in the retina results in a slowly changing edge, as in (a), being sharpened, as in (b). This makes it easier to separate objects in the image, but produces an optical illusion called *Mach bands*. Near the edge, the overshoot makes the dark region look darker, and the light region look lighter. This produces dark and light bands that run parallel to the edge.

Humans and other animals use vision to identify nearby objects, such as enemies, food, and mates. This is done by distinguishing one region in the image from another, based on differences in brightness and color. In other words, the first step in recognizing an object is to identify its *edges*, the discontinuity that separates an object from its background. The middle layer of the retina helps this task by sharpening the edges in the viewed image. As an illustration of how this works, Fig. 24-2 shows an image that slowly changes from dark to light, producing a blurry and poorly defined edge. Figure (a) shows the intensity profile of this image, the pattern of brightness entering the eye. Figure (b) shows the brightness profile appearing on the optic nerve, the image transmitted to the brain. The processing in the retina makes the edge between the light and dark areas appear more abrupt, reinforcing that the two regions are different.

The overshoot in the edge response creates an interesting optical illusion. Next to the edge, the dark region appears to be unusually dark, and the light region appears to be unusually light. The resulting light and dark strips are called **Mach bands**, after Ernst Mach (1838-1916), an Austrian physicist who first described them.

As with one-dimensional signals, image convolution can be viewed in two ways: from the input, and from the output. From the input side, each pixel in